Simultaneous Localization and Mapping
Introduction and EKF-SLAM

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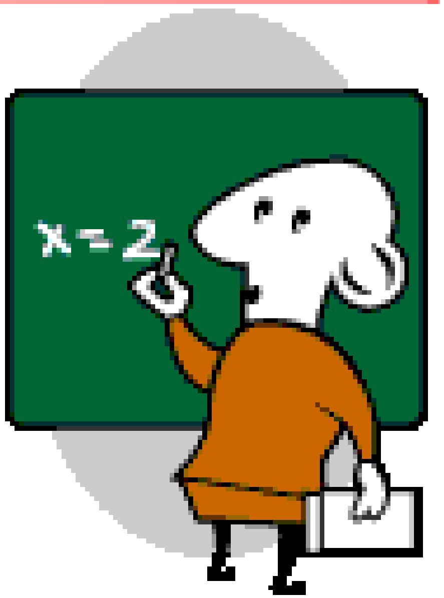
Lecture slides heavily use material from the textbook and
Sebastian Thrun, Lecture Slides; http://www.probabilistic-robotics.org/

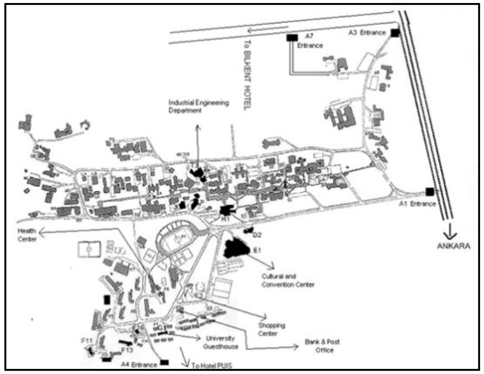
What we will discuss



- Define the SLAM problem,
- Discuss why it is more difficult that anything we have seen so far,
- Discuss On-line SLAM and Full SLAM problems and techniques to generate consistent maps,
- Briefly discuss Map-Matching,
- Apply Kalman Filters to SLAM: EKF-SLAM,
- Focus on Landmark based case,
- Discuss the known correspondence case,
- Extend it to unknown correspondence case,
- Discuss main limitations,
- Discuss practical issues to increase efficiency,
- Summarize

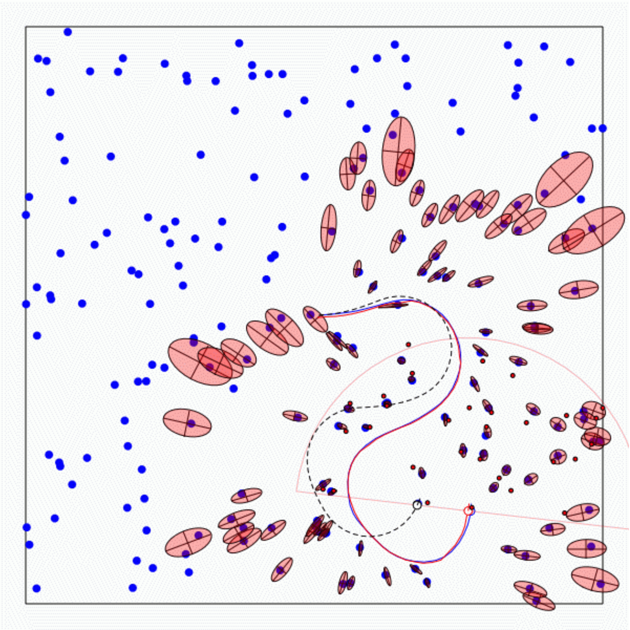
The SLAM Problem

- **A fundamental problem of autonomous mobile systems,**
- **Called "*Simultaneous Localization and Mapping*"-SLAM**
- **OR: "*Concurrent Mapping and Localization*" – CML,**
- **No access to map of the environment, no knowledge of location: Both to be extracted from sensor observations,**
- ***"Move around, acquire a map of the environment and localize yourself relative to this map, correct the map as you gain more information"***
- ***Humans do it all the time*!!**
- **(Remember your first days at METU)**



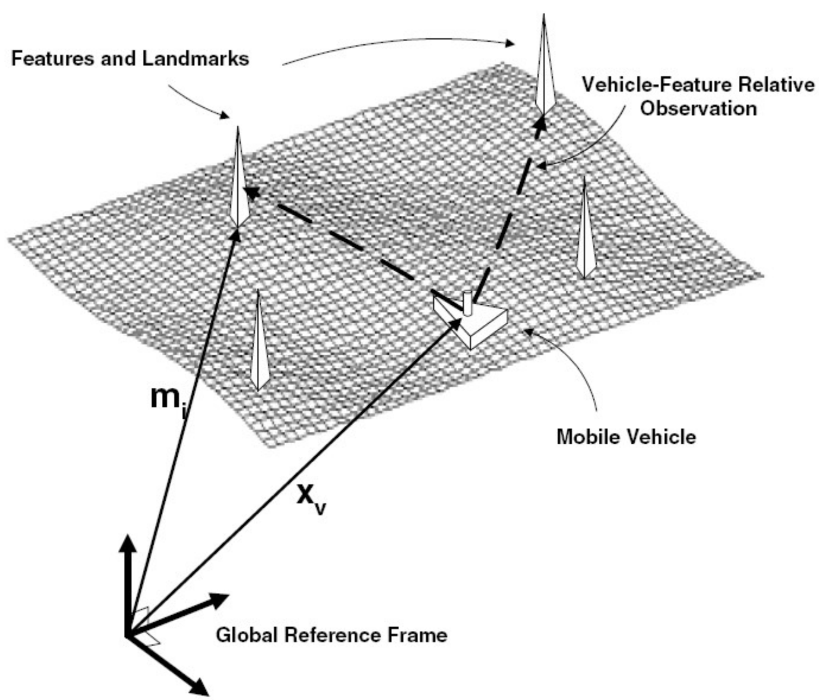
The SLAM Problem

- A robot is exploring an unknown, static environment,
- GIVEN:
  - The robot's controls,
  - Measurements of nearby features,
- NEEDED:
  - Map of features (Locations w.r.t a relative coordinate frame based on initial location),
  - Trajectory of the robot



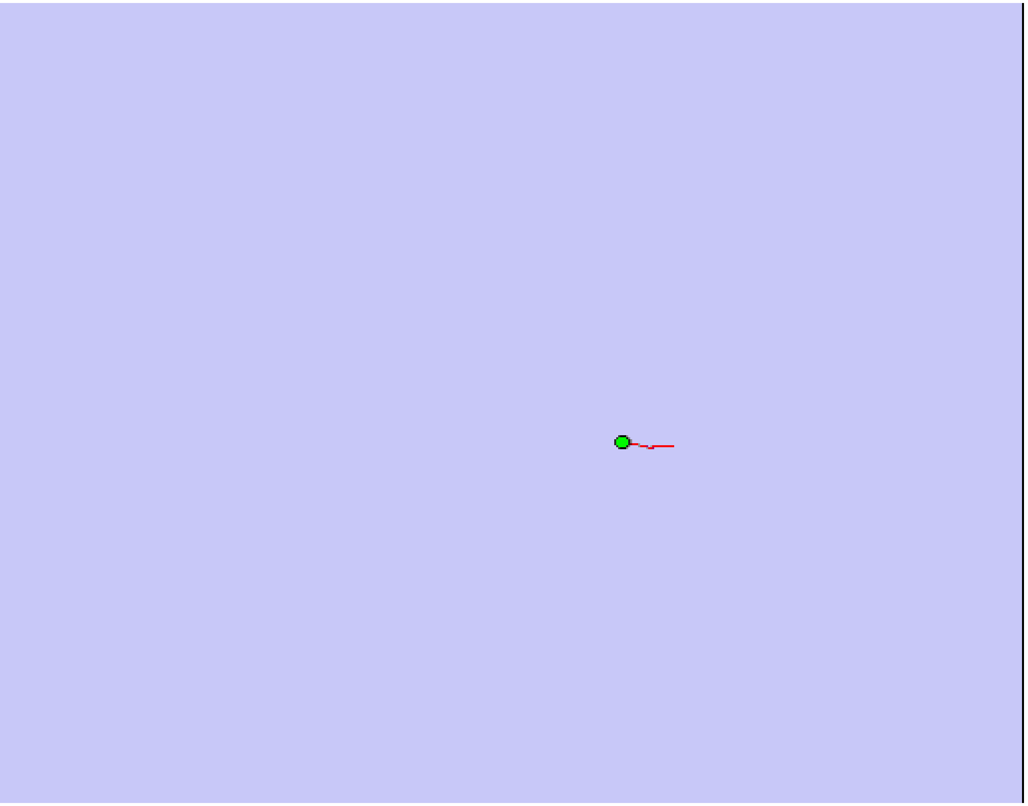
Structure of Landmark Based SLAM Problem

- We assume point landmarks rather than raw measurements,
- Sensor provides range and bearing info (may also provide bearing only! E.g. monocular Camera)



Why is SLAM needed?

- **Assume you generate a map based purely on Odometry information (crude estimate of where you are using wheel encoders or Inertial Measurement Units)**

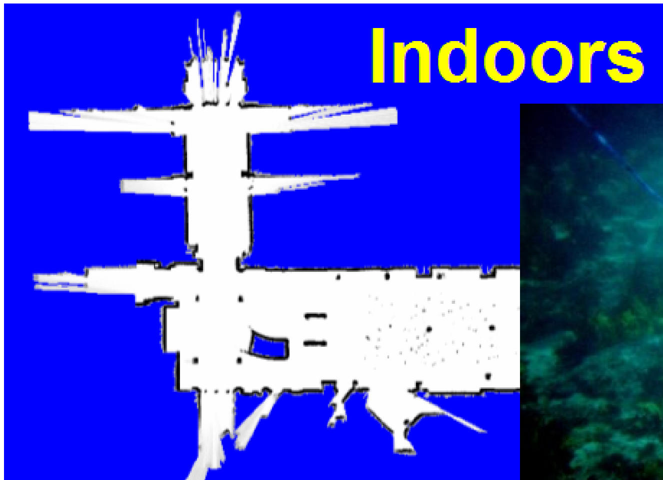


Mapping the Intel Lab: Raw Odometry

Video: http://robots.stanford.edu/videos.html

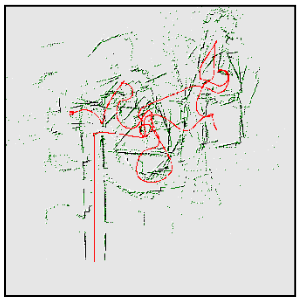
Why is SLAM Important?

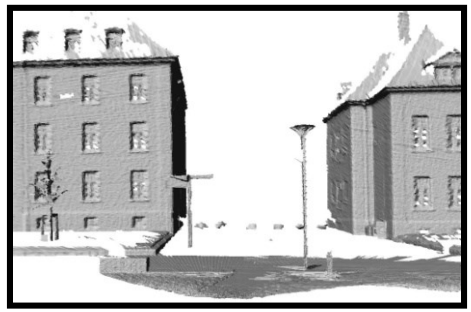
SLAM Applications

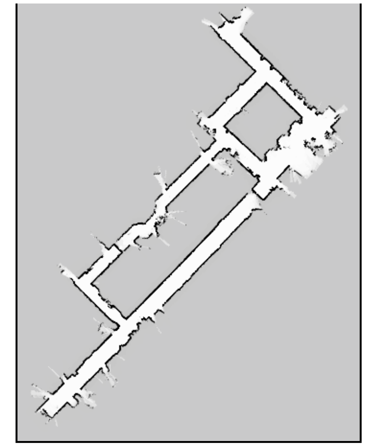


Representations of Maps

- Grid maps or scans

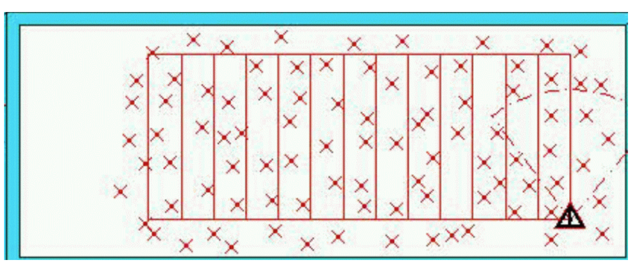


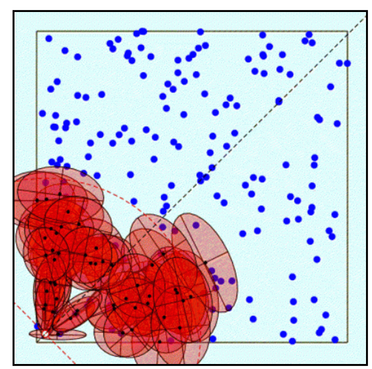


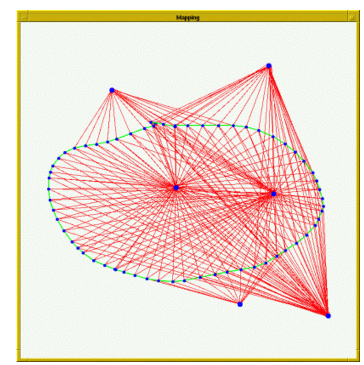


[Lu & Milios, 97; Gutmann, 98: Thrun 98; Burgard, 99; Konolige & Gutmann, 00; Thrun, 00; Arras, 99; Haehnel, 01;…]

- Landmark-based



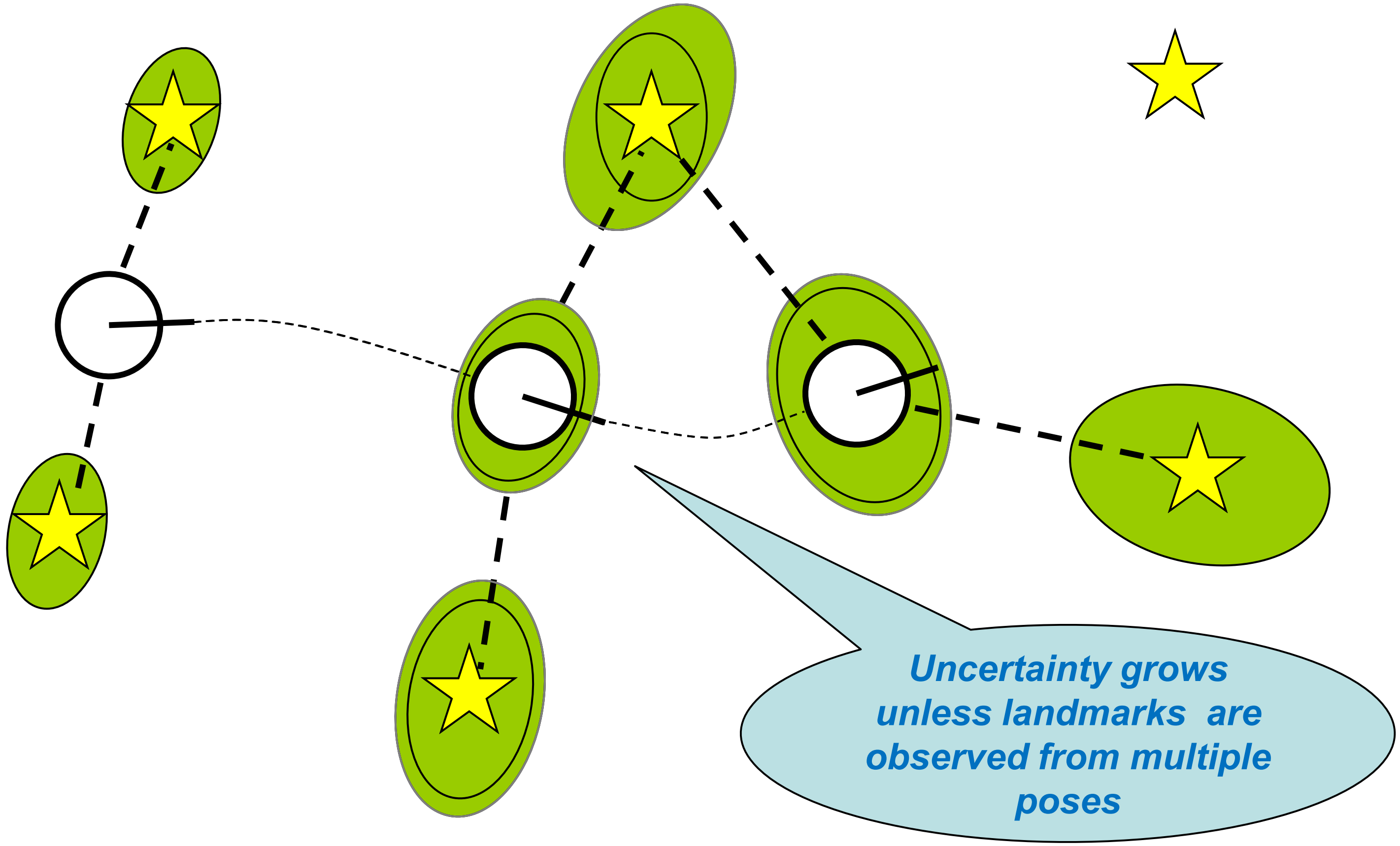




[Leonard et al., 98; Castelanos et al., 99: Dissanayake et al., 2001; Montemerlo et al., 2002;…

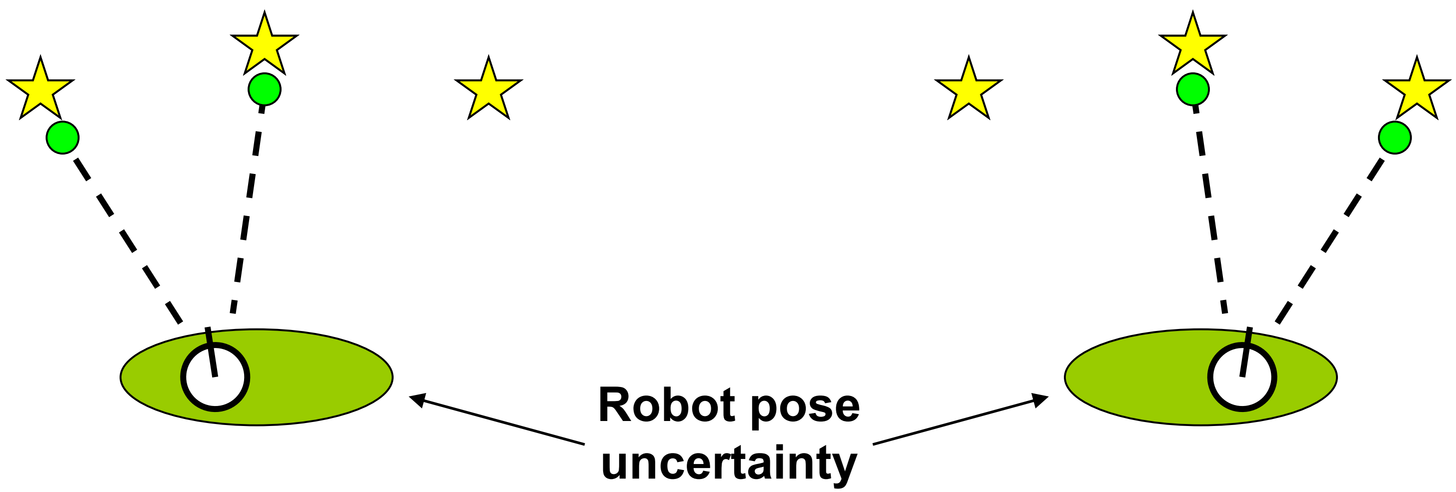
Why is SLAM a harder problem?

SLAM: robot path and map are both unknown



Robot path error correlates errors in the map

Why is SLAM a harder problem?



- In the real world, the mapping between observations and landmark IDs is often unknown
- Picking wrong data associations can have severe consequences (*performance* and *convergence*)
- Pose error correlates measurements and data associations

Primary forms of SLAM Problem

- ***Full SLAM:***

Estimates entire path and map!

$$p(x_{1:t}, m \mid z_{1:t}, u_{1:t})$$

- ***Online SLAM:***

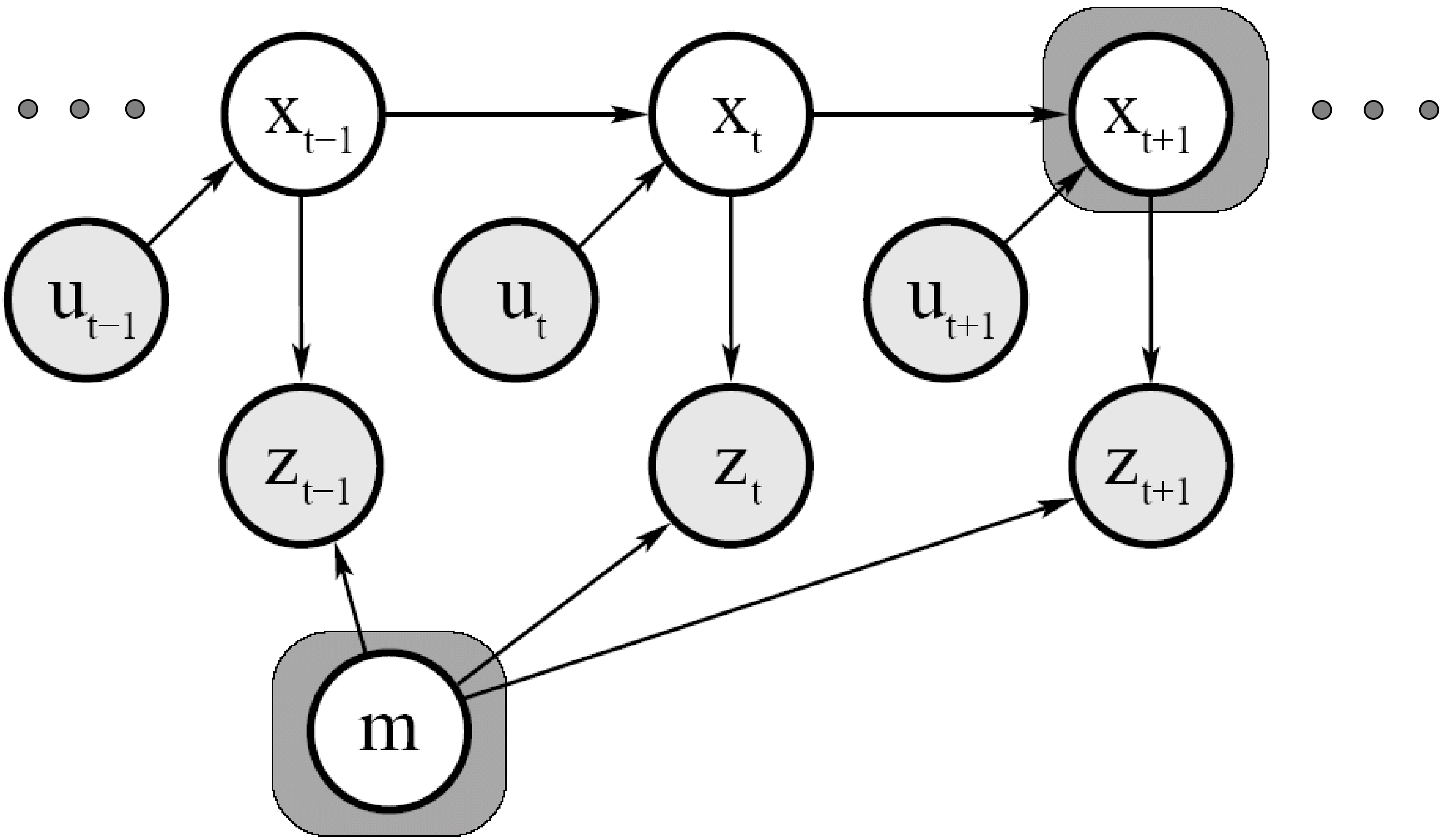
This is a marginal density

$$p(x_t, m \mid z_{1:t}, u_{1:t}) = \int \int \dots \int p(x_{1:t}, m \mid z_{1:t}, u_{1:t})\, dx_1 dx_2 \dots dx_{t-1}$$

Integrations typically done recursively one at a time

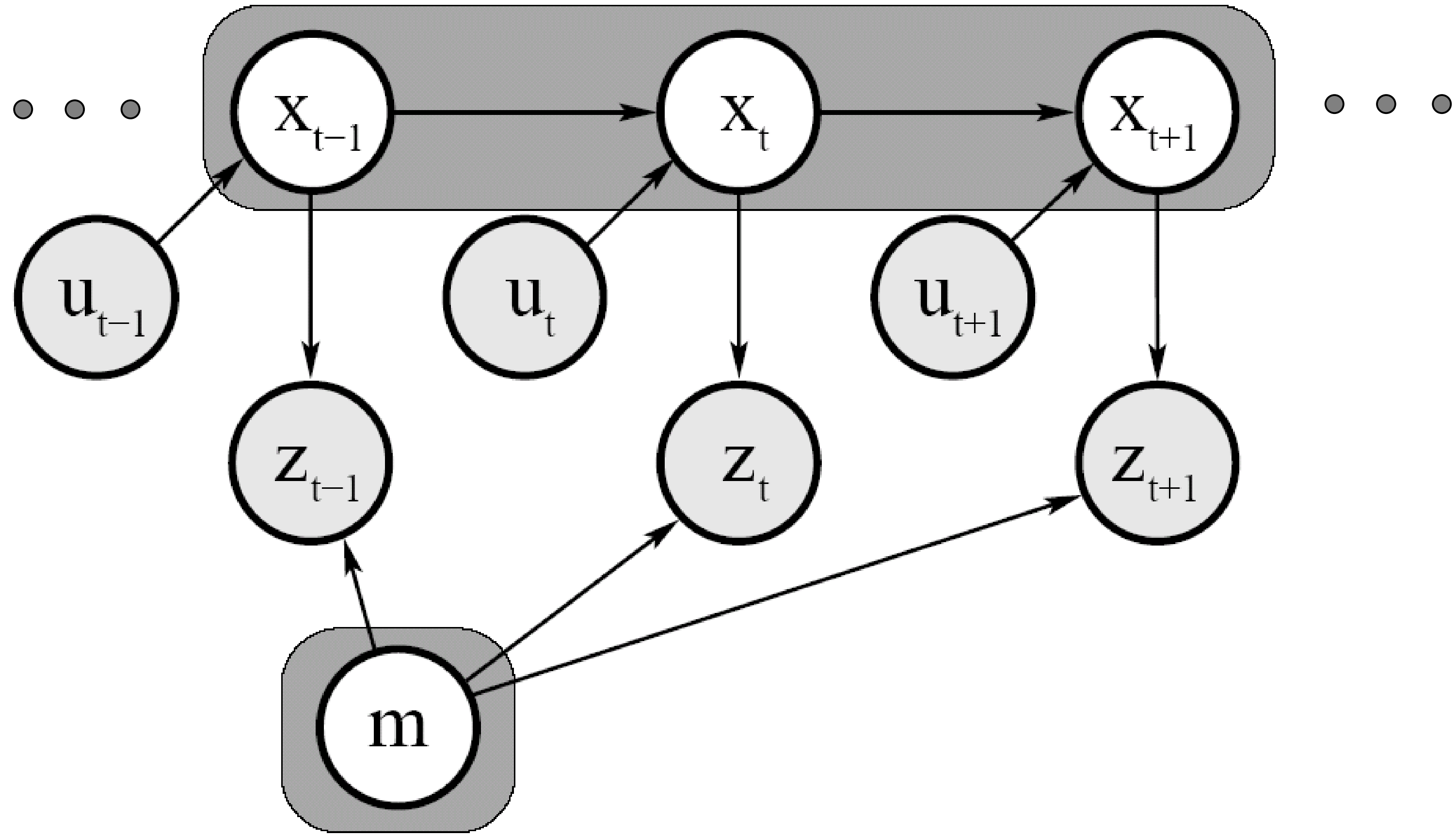
Estimates most recent pose and map!

Graph (network) model of On-Line SLAM



$$p(x_t, m \mid z_{1:t}, u_{1:t}) = \int\int \ldots \int p(x_{1:t}, m \mid z_{1:t}, u_{1:t})\, dx_1 dx_2 \ldots dx_{t-1}$$

Graph (network) model of Full SLAM



$$p(x_{1:t}, m \mid z_{1:t}, u_{1:t})$$

Generating Consistent Maps

- **Scan matching**
- **EKF SLAM**
- **Fast-SLAM**
- **Graph-SLAM, SEIFs (*spa*… *ed***

Simple maximum likelihood matching

Our primary focus today – Maybe the earliest and most influential

Particle Filters and EKF Hybrid – Later!

Off-Line Full-SLAM, Stores constraint data and performs Inference off-line (outside of current scope)

Scan Matching

The basic idea:

- One tries to find the incremental robot motion and build the map by:
  - Get a range-scan at every step (a 2D point cloud),
  - Try to *match it* with the scan at the previous step,
  - Find the robot translation and rotation that gives the *best match.* Hence find the new robot location,
  - Also build a map by using (merging) range-scans and the robot pose computed as a *known pose*.

- Now let us try to express it more formally:

Scan Matching

Maximize the likelihood of the *i*-th pose and map relative to the (*i-1*)-th pose and map.

$$\hat{x}_t = \underset{x_t}{\arg\max} \left\{ p(z_t \mid x_t, \hat{m}^{[t-1]}) \cdot p(x_t \mid u_{t-1}, \hat{x}_{t-1}) \right\}$$

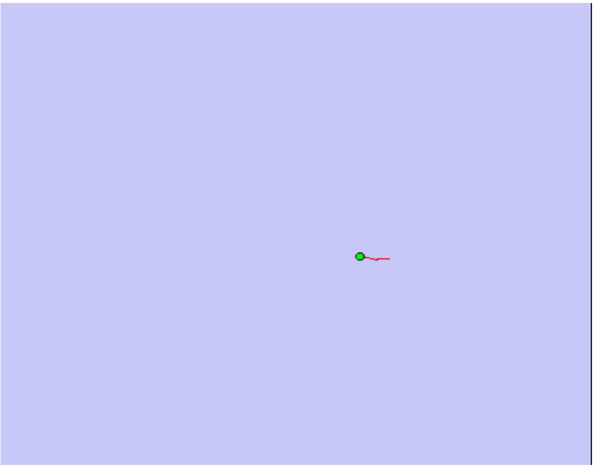
current measurement

robot motion

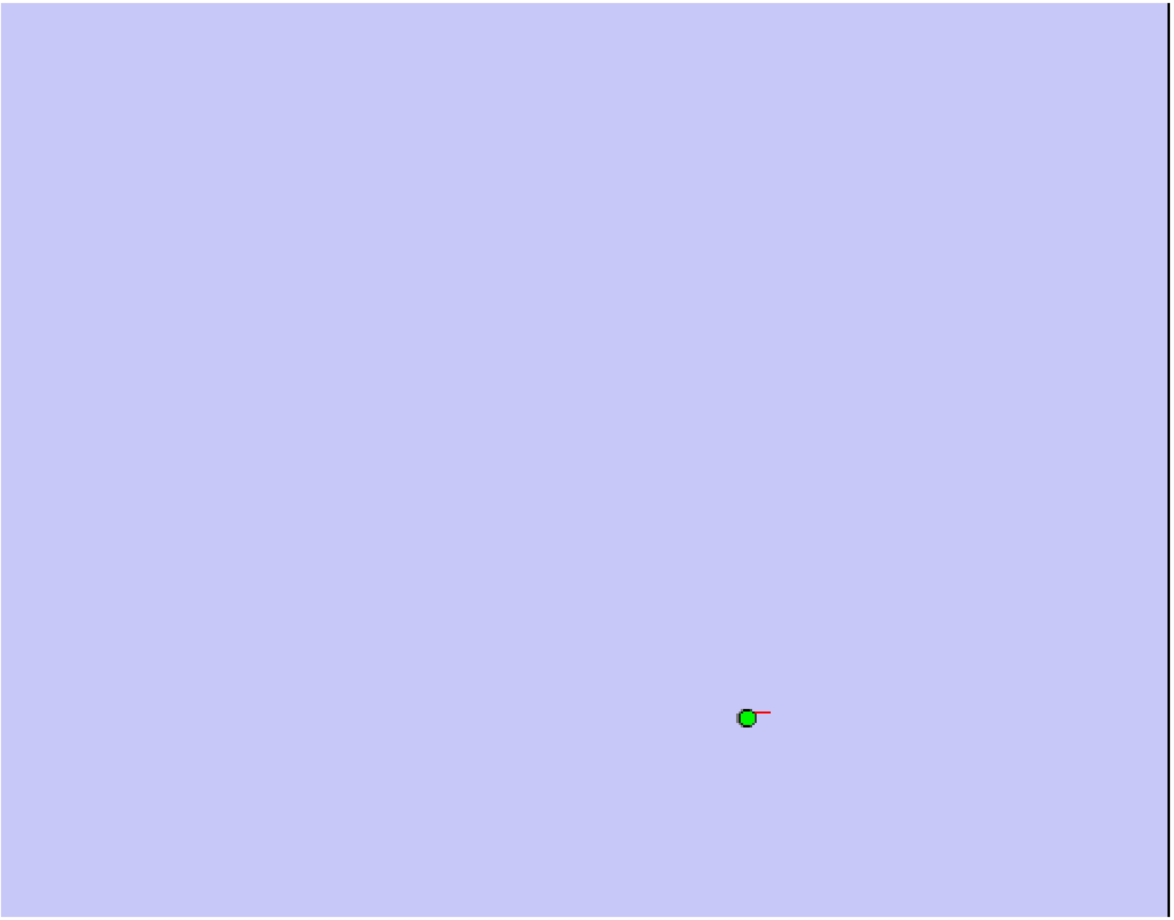
map constructed so far

Then calculate the map $\hat{m}^{[t]}$ according to "mapping with known poses" based on the poses and measurements (range-scans). (e.g. occupancy grid mapping…)

Illustration - Scan Matching

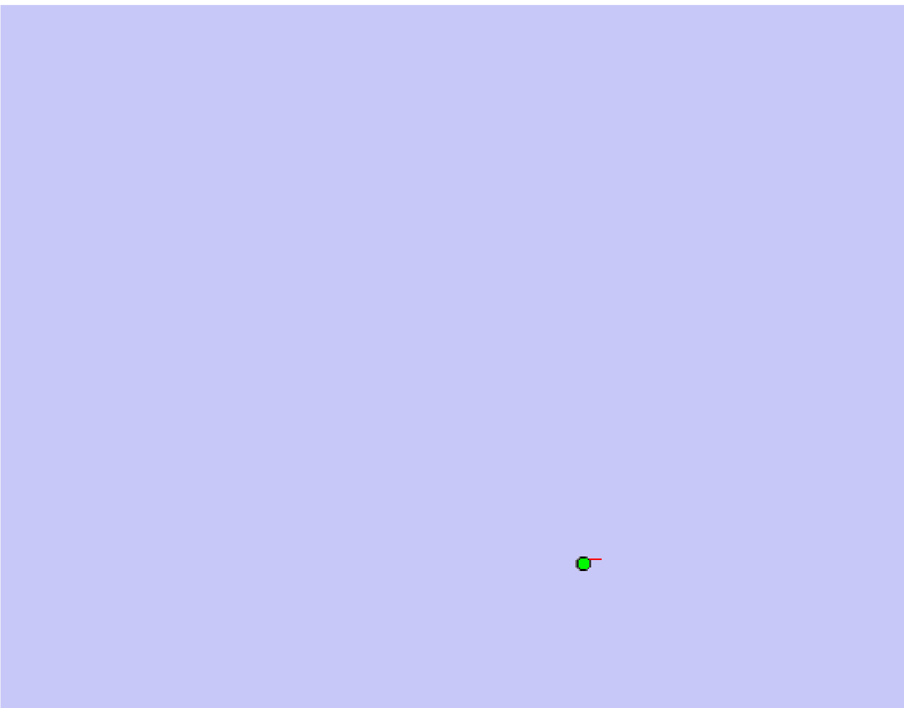


Mapping the Intel Lab: With odometry

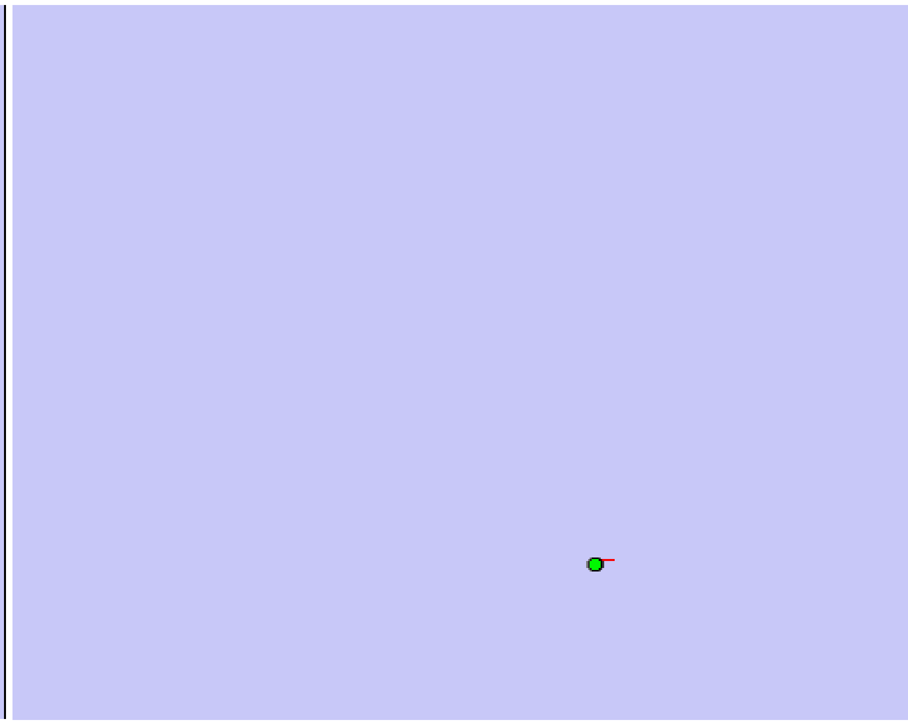


Mapping the Intel Lab: Scan Matching

Compare with SLAM (FastSLAM)



Mapping the Intel Lab: Scan Matching



Mapping the Intel Lab: FastSLAM

What differences in behavior do you see?

Scan Matching Assessment

- Significantly better than odometry only,
- Still accumulates relative pose information (hence building up of errors on localization),
- Can correct pose w.r.t revisited map locations (If scan matching is done against a global map),
- But cannot propagate pose corrections (hence map corrections) backward to previous pose steps.
- Bayes Filter based SLAM algorithms are able to do this...

(Extended) Kalman Filtering Algorithm

1. **Extended_Kalman_filter**(μ_{t-1}, Σ_{t-1}, u_t, z_t):

2. Prediction:
3. $\bar{\mu}_t = g(u_t, \mu_{t-1})$ $\longleftarrow$ $\bar{\mu}_t = A_t \mu_{t-1} + B_t u_t$
4. $\bar{\Sigma}_t = G_t \Sigma_{t-1} G_t^T + R_t$ $\longleftarrow$ $\bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + R_t$

5. Correction:
6. $K_t = \bar{\Sigma}_t H_t^T (H_t \bar{\Sigma}_t H_t^T + Q_t)^{-1}$ $\longleftarrow$ $K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1}$
7. $\mu_t = \bar{\mu}_t + K_t (z_t - h(\bar{\mu}_t))$ $\longleftarrow$ $\mu_t = \bar{\mu}_t + K_t (z_t - C_t \bar{\mu}_t)$
8. $\Sigma_t = (I - K_t H_t) \bar{\Sigma}_t$ $\longleftarrow$ $\Sigma_t = (I - K_t C_t) \bar{\Sigma}_t$

9. Return μ_t, Σ_t

$$G_t = \left. \frac{\partial g(u_t, x_{t-1})}{\partial x_{t-1}} \right|_{x_{t-1} = \mu_{t-1}} \qquad H_t = \left. \frac{\partial h(x_t)}{\partial x_t} \right|_{x_t = \bar{\mu}_t}$$

Belief in (E)KF-SLAM

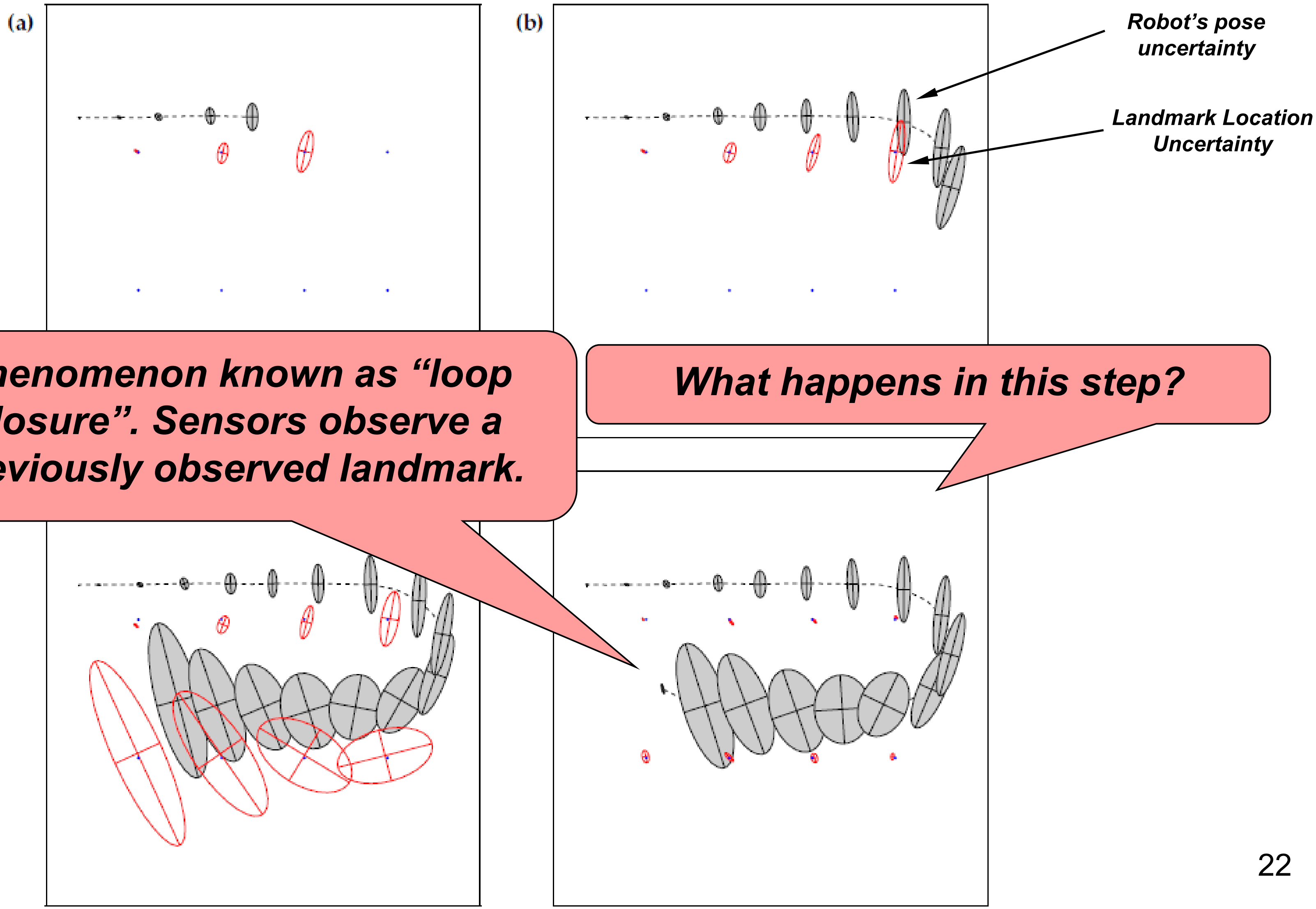
- **Pose and Map with N landmarks represented by: (3+2N)-dimensional Gaussian**

$$Bel(y_t) = Bel(x_t, m_t) = \left\langle \begin{pmatrix} \bar{x} \\ \bar{y} \\ \bar{\theta} \\ \bar{l}_1 \\ \bar{l}_2 \\ \vdots \\ \bar{l}_N \end{pmatrix}, \begin{pmatrix} \sigma_x^2 & \sigma_{xy} & \sigma_{x\theta} & \sigma_{xl_1} & \sigma_{xl_2} & \cdots & \sigma_{xl_N} \\ \sigma_{xy} & \sigma_y^2 & \sigma_{y\theta} & \sigma_{yl_1} & \sigma_{yl_2} & \cdots & \sigma_{yl_N} \\ \sigma_{x\theta} & \sigma_{y\theta} & \sigma_\theta^2 & \sigma_{\theta l_1} & \sigma_{\theta l_2} & \cdots & \sigma_{\theta l_N} \\ \sigma_{xl_1} & \sigma_{yl_1} & \sigma_{\theta l_1} & \sigma_{l_1}^2 & \sigma_{l_1 l_2} & \cdots & \sigma_{l_1 l_N} \\ \sigma_{xl_2} & \sigma_{yl_2} & \sigma_{\theta l_2} & \sigma_{l_1 l_2} & \sigma_{l_2}^2 & \cdots & \sigma_{l_2 l_N} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{xl_N} & \sigma_{yl_N} & \sigma_{\theta l_N} & \sigma_{l_1 l_N} & \sigma_{l_2 l_N} & \cdots & \sigma_{l_N}^2 \end{pmatrix} \right\rangle$$

Each landmark l_i position consist of x_i, y_i

- **Assumes landmark based map and Gaussian noise**
- **State augmented by landmark variables,**
- **Can handle hundreds of dimensions**

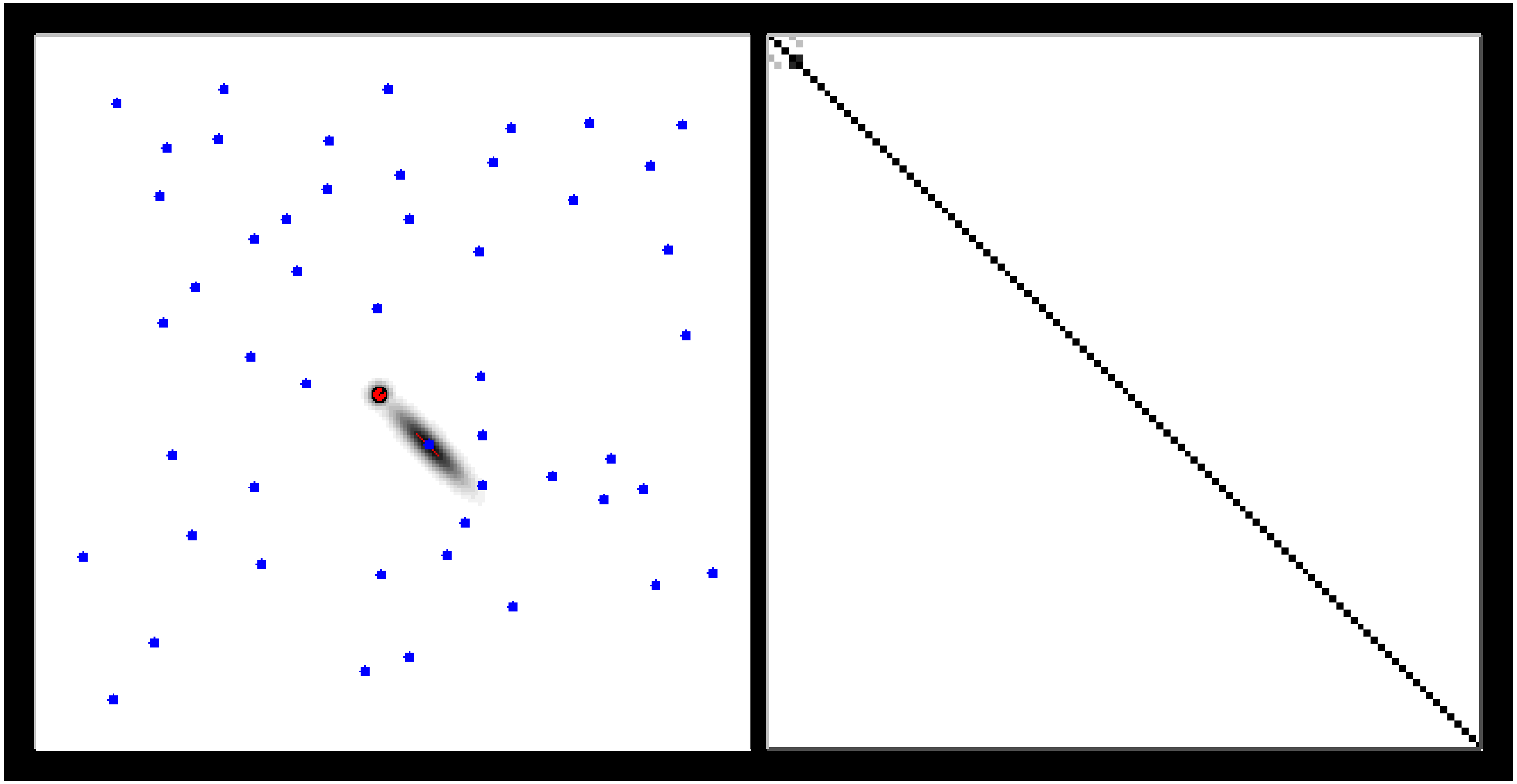
EKF-SLAM Simple Example



Phenomenon known as “loop closure”. Sensors observe a previously observed landmark.

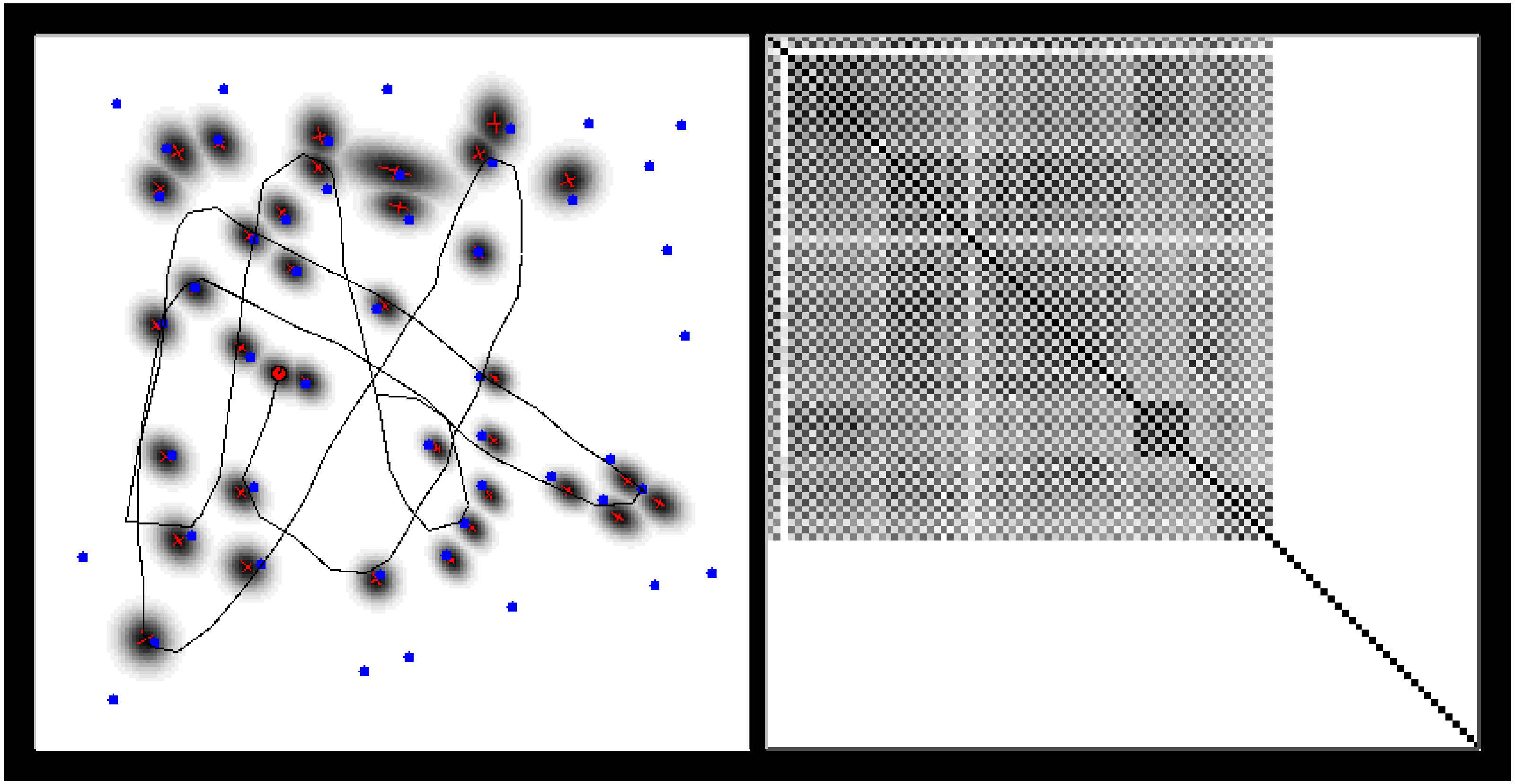
What happens in this step?

EKF-SLAM Another Simple Example



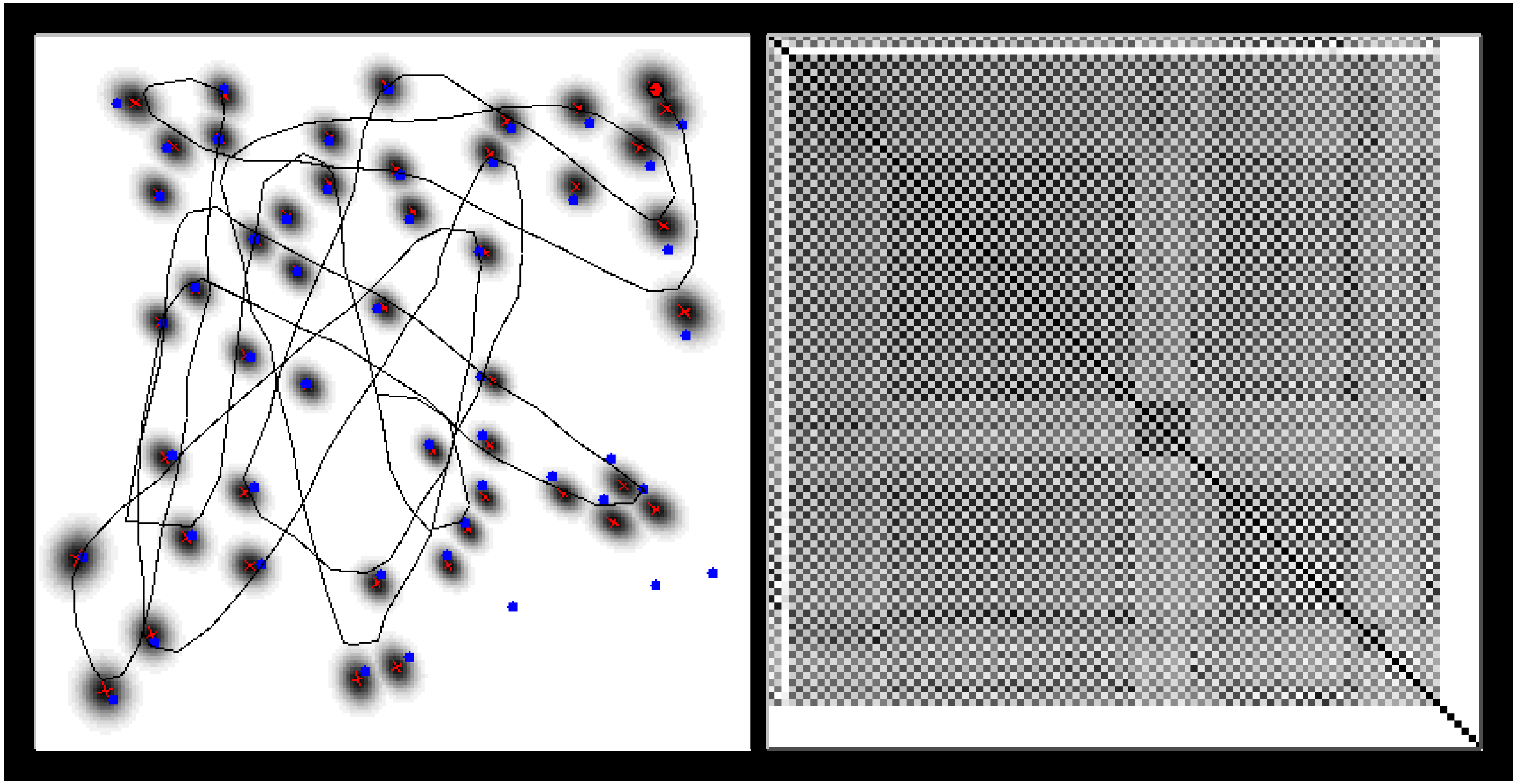
Map *Covariance matrix*

EKF-SLAM “Correlations”



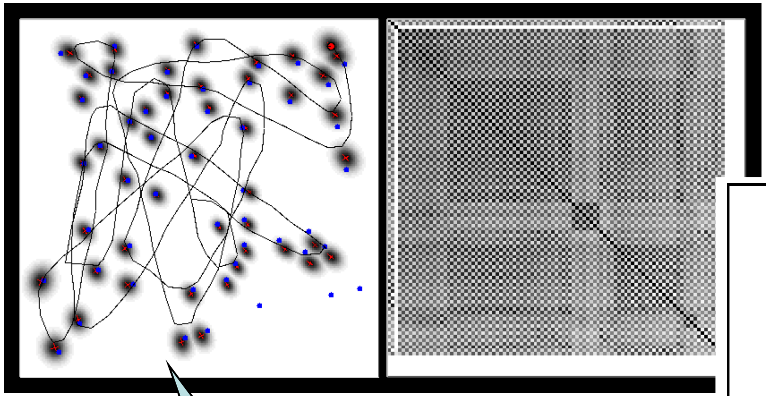
Map *Covariance matrix*

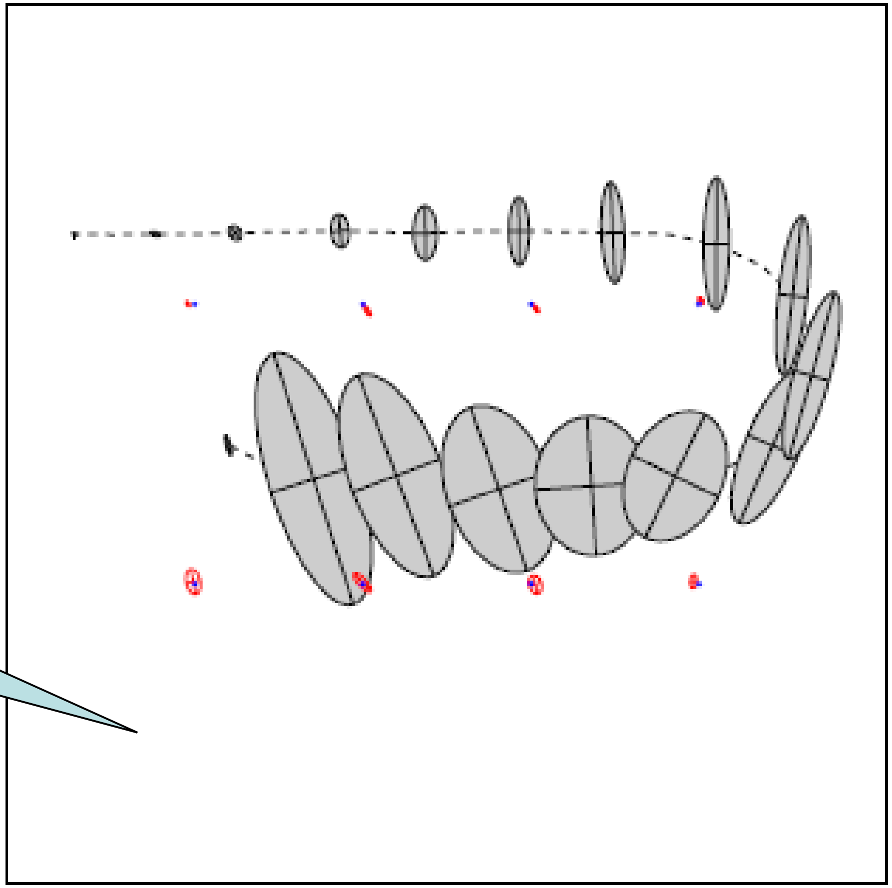
EKF-SLAM Another Simple Example



Map *Covariance matrix*

Covariance Matrix





What is not visible in these figures?

Convergence of KF-SLAM (Linear)

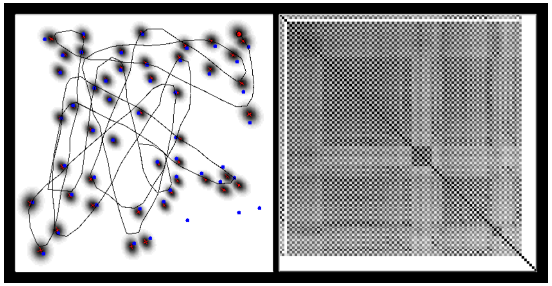
[Dissanayake et al., 2001]

***Theorem*:**

The determinant of any sub-matrix of the map covariance matrix decreases monotonically as successive observations are made.

***Theorem*:**

In the limit the landmark estimates become fully correlated



Derivation of EKF-SLAM - 1

Known correspondence case

- **The Motion Model and Linearization**

$$y_t = y_{t-1} + \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t\Delta t + \gamma_t\Delta t \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$y_t = y_{t-1} + \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega_t\Delta t + \gamma_t\Delta t \end{pmatrix}$$

$$F_x = \begin{pmatrix} 1 & 0 & 0 & 0 \cdots 0 \\ 0 & 1 & 0 & 0 \cdots 0 \\ 0 & 0 & 1 & \underbrace{0 \cdots 0}_{3N \text{ columns}} \end{pmatrix}$$

I believe this should not be here!!
(this is stated as the noise-free velocity motion model)

$$y_t = \underbrace{y_{t-1} + F_x^T \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta+\omega_t\Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta+\omega_t\Delta t) \\ \omega\Delta t \end{pmatrix}}_{g(u_t, y_{t-1})} + \mathcal{N}(0, F_x^T\, R_t\, F_x)$$

Full Motion model with Noise

Derivation of EKF-SLAM - 2

- **The Motion Model and Linearization**

$$G_t = \left.\frac{\partial g(u_t, y_{t-1})}{\partial y_{t-1}}\right|_{x_{t-1}=\mu_{t-1}}$$

$$g(u_t, y_{t-1}) \approx g(u_t, \mu_{t-1}) + G_t\,(y_{t-1} - \mu_{t-1})$$

$$y_t = \underbrace{y_{t-1} + F_x^T \begin{pmatrix} -\frac{v_t}{\omega_t}\sin\theta + \frac{v_t}{\omega_t}\sin(\theta + \omega_t \Delta t) \\ \frac{v_t}{\omega_t}\cos\theta - \frac{v_t}{\omega_t}\cos(\theta + \omega_t \Delta t) \\ \omega \Delta t \end{pmatrix}}_{g(u_t, y_{t-1})} + \mathcal{N}(0, F_x^T\, R_t\, F_x)$$

Jacobian of state transition function g(.)

$$G_t = I + F_x^T\, g_t\, F_x$$

Jacobian is sparse and can be expressed in terms of a low-dim Jacobian g_t

$$g_t = \begin{pmatrix} 0 & 0 & -\frac{v_t}{\omega_t}\cos\mu_{t-1,\theta} + \frac{v_t}{\omega_t}\cos(\mu_{t-1,\theta} + \omega_t \Delta t) \\ 0 & 0 & -\frac{v_t}{\omega_t}\sin\mu_{t-1,\theta} + \frac{v_t}{\omega_t}\sin(\mu_{t-1,\theta} + \omega_t \Delta t) \\ 0 & 0 & 0 \end{pmatrix}$$

Derivation of EKF-SLAM - 3

- **The Measurement Model and Linearization**

$$z_t^i = \underbrace{\begin{pmatrix} \sqrt{(m_{j,x}-x)^2+(m_{j,y}-y)^2} \\ \operatorname{atan2}(m_{j,y}-y,\, m_{j,x}-x)-\theta \\ m_{j,s} \end{pmatrix}}_{h(y_t,j)} + \mathcal{N}(0, \underbrace{\begin{pmatrix} \sigma_r & 0 & 0 \\ 0 & \sigma_\phi & 0 \\ 0 & 0 & \sigma_s \end{pmatrix}}_{Q_t})$$

$$H_t = \left.\frac{\partial h(y_t)}{\partial y_t}\right|_{x_t=\bar{\mu}_t}$$

$$h(y_t,j) \approx h(\bar{\mu}_t,j) + \quad (y_t-\bar{\mu}_t)$$

Again: Taylor series approximation of h(.)

$$H_t^i = h_t^i\, F_{x,j}$$

$$h_t^i = \begin{pmatrix} \frac{\bar{\mu}_{t,x}-\bar{\mu}_{j,x}}{\sqrt{q_t}} & \frac{\bar{\mu}_{t,y}-\bar{\mu}_{j,y}}{\sqrt{q_t}} & 0 & \frac{\bar{\mu}_{j,x}-\bar{\mu}_{t,x}}{\sqrt{q_t}} & \frac{\bar{\mu}_{j,y}-\bar{\mu}_{t,y}}{\sqrt{q_t}} & 0 \\ \frac{\bar{\mu}_{j,y}-\bar{\mu}_{t,y}}{q_t} & \frac{\bar{\mu}_{t,x}-\bar{\mu}_{j,x}}{q_t} & -1 & \frac{\bar{\mu}_{t,y}-\bar{\mu}_{j,y}}{q_t} & \frac{\bar{\mu}_{j,x}-\bar{\mu}_{t,x}}{q_t} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$3 \times (3N+3)$$

$$F_{x,j} = \begin{pmatrix} 1 & 0 & 0 & 0\cdots0 & 0 & 0 & 0 & 0\cdots0 \\ 0 & 1 & 0 & 0\cdots0 & 0 & 0 & 0 & 0\cdots0 \\ 0 & 0 & 1 & 0\cdots0 & 0 & 0 & 0 & 0\cdots0 \\ 0 & 0 & 0 & 0\cdots0 & 1 & 0 & 0 & 0\cdots0 \\ 0 & 0 & 0 & 0\cdots0 & 0 & 1 & 0 & 0\cdots0 \\ 0 & 0 & 0 & \underbrace{0\cdots0}_{3j-3} & 0 & 0 & 1 & \underbrace{0\cdots0}_{3N-3j} \end{pmatrix}$$

Derivation of EKF-SLAM - 4

- How should we initialize the mean and variance of newly seen landmarks?
- A bad idea:

$$\mu_0 = (\ 0\ 0\ 0\ \ldots\ 0\ )^T$$

$$\Sigma_0 = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \infty & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \infty \end{pmatrix}$$

Bad, because the first measurement of the landmark falls far away from the mean. What happens then?

- A better idea is to initialize based on the first measurement made from the new landmark:

$$\begin{pmatrix} \bar{\mu}_{j,x} \\ \bar{\mu}_{j,y} \\ \bar{\mu}_{j,s} \end{pmatrix} = \begin{pmatrix} \bar{\mu}_{t,x} \\ \bar{\mu}_{t,y} \\ s_t^i \end{pmatrix} + \begin{pmatrix} r_t^i \cos(\phi_t^i + \bar{\mu}_{t,\theta}) \\ r_t^i \sin(\phi_t^i + \bar{\mu}_{t,\theta}) \\ 0 \end{pmatrix}$$

And how can I initialize the covariance?

- Possible since measurement model is bijective.

Derivation of EKF-SLAM - 5

- Covariance can be initialized based on a series of observations of the new landmark.
- If the measurement model in not bijective (*one-to-one* and *onto*), then again, multiple measurements are necessary to initialize the mean in a meaningful way.

- Question: Give me an example application where a non-bijective measurement is made??

EKF-SLAM Algorithm – 1

Algorithm EKF_SLAM_known_correspondences$(\mu_{t-1}, \Sigma_{t-1}, u_t, z_t, c_t)$:

$$F_x = \begin{pmatrix} 1 & 0 & 0 & 0 \cdots 0 \\ 0 & 1 & 0 & 0 \cdots 0 \\ 0 & 0 & 1 & \underbrace{0 \cdots 0}_{3N} \end{pmatrix}$$

$$\bar{\mu}_t = \mu_{t-1} + F_x^T \begin{pmatrix} -\frac{v_t}{\omega_t} \sin \mu_{t-1,\theta} + \frac{v_t}{\omega_t} \sin(\mu_{t-1,\theta} + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos \mu_{t-1,\theta} - \frac{v_t}{\omega_t} \cos(\mu_{t-1,\theta} + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix}$$

$$G_t = I + F_x^T \begin{pmatrix} 0 & 0 & -\frac{v_t}{\omega_t} \cos \mu_{t-1,\theta} + \frac{v_t}{\omega_t} \cos(\mu_{t-1,\theta} + \omega_t \Delta t) \\ 0 & 0 & -\frac{v_t}{\omega_t} \sin \mu_{t-1,\theta} + \frac{v_t}{\omega_t} \sin(\mu_{t-1,\theta} + \omega_t \Delta t) \\ 0 & 0 & 0 \end{pmatrix} F_x$$

$$\bar{\Sigma}_t = G_t \, \Sigma_{t-1} \, G_t^T + F_x^T \, R_t \, F_x$$

$$Q_t = \begin{pmatrix} \sigma_r & 0 & 0 \\ 0 & \sigma_\phi & 0 \\ 0 & 0 & \sigma_s \end{pmatrix}$$

for all observed features $z_t^i = (r_t^i \ \ \phi_t^i \ \ s_t^i)^T$ *do*

$j = c_t^i$

if landmark j *never seen before*

$$\begin{pmatrix} \bar{\mu}_{j,x} \\ \bar{\mu}_{j,y} \\ \bar{\mu}_{j,s} \end{pmatrix} = \begin{pmatrix} \bar{\mu}_{t,x} \\ \bar{\mu}_{t,y} \\ s_t^i \end{pmatrix} + \begin{pmatrix} r_t^i \ \cos(\phi_t^i + \bar{\mu}_{t,\theta}) \\ r_t^i \ \sin(\phi_t^i + \bar{\mu}_{t,\theta}) \\ 0 \end{pmatrix}$$

endif

$$\delta = \begin{pmatrix} \delta_x \\ \delta_y \end{pmatrix} = \begin{pmatrix} \bar{\mu}_{j,x} - \bar{\mu}_{t,x} \\ \bar{\mu}_{j,y} - \bar{\mu}_{t,y} \end{pmatrix}$$

$q = \delta^T \delta$

EKF-SLAM Algorithm – 2

$$\hat{z}_t^i = \begin{pmatrix} \sqrt{q} \\ \operatorname{atan2}(\delta_y, \delta_x) - \bar{\mu}_{t,\theta} \\ \bar{\mu}_{j,s} \end{pmatrix}$$

$$F_{x,j} = \begin{pmatrix} 1 & 0 & 0 & 0 \cdots 0 & 0 & 0 & 0 & 0 \cdots 0 \\ 0 & 1 & 0 & 0 \cdots 0 & 0 & 0 & 0 & 0 \cdots 0 \\ 0 & 0 & 1 & 0 \cdots 0 & 0 & 0 & 0 & 0 \cdots 0 \\ 0 & 0 & 0 & 0 \cdots 0 & 1 & 0 & 0 & 0 \cdots 0 \\ 0 & 0 & 0 & 0 \cdots 0 & 0 & 1 & 0 & 0 \cdots 0 \\ 0 & 0 & 0 & \underbrace{0 \cdots 0}_{3j-3} & 0 & 0 & 1 & \underbrace{0 \cdots 0}_{3N-3j} \end{pmatrix}$$

$$H_t^i = \frac{1}{q} \begin{pmatrix} \sqrt{q}\delta_x & -\sqrt{q}\delta_y & 0 & -\sqrt{q}\delta_x & \sqrt{q}\delta_y & 0 \\ \delta_y & \delta_x & -1 & -\delta_y & -\delta_x & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} F_{x,j}$$

$$K_t^i = \bar{\Sigma}_t \; H_t^{iT} (H_t^i \; \bar{\Sigma}_t \; H_t^{iT} + Q_t)^{-1}$$

$$\bar{\mu}_t = \bar{\mu}_t + K_t^i (z_t^i - \hat{z}_t^i)$$

$$\bar{\Sigma}_t = (I - K_t^i \; H_t^i) \; \bar{\Sigma}_t$$

endfor

$\mu_t = \bar{\mu}_t$

$\Sigma_t = \bar{\Sigma}_t$

return μ_t, Σ_t

EKF-SLAM Unknown Correspondence

- Simplest approach is the maximum likelihood data association.
- Measurement log-likelihood is computed for each N existing landmark as well as for an $(N+1)^{st}$ hypothesis of a new landmark,
- Landmark maximizing the likelihood chosen,
- If all likelihoods below a certain threshold, a new $(N+1)^{st}$ landmark is created,
- Very similar to EKF localization with unknown correspondence already discussed,
- Only the measurement update cycle is modified

EKF-SLAM Unknown Correspondence

Algorithm EKF_SLAM($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t, N_{t-1}$):

$$N_t = N_{t-1}$$

$$F_x = \begin{pmatrix} 1 & 0 & 0 & 0 \cdots 0 \\ 0 & 1 & 0 & 0 \cdots 0 \\ 0 & 0 & 1 & 0 \cdots 0 \end{pmatrix}$$

$$\bar{\mu}_t = \mu_{t-1} + F_x^T \begin{pmatrix} -\frac{v_t}{\omega_t} \sin \mu_{t-1,\theta} + \frac{v_t}{\omega_t} \sin(\mu_{t-1,\theta} + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos \mu_{t-1,\theta} - \frac{v_t}{\omega_t} \cos(\mu_{t-1,\theta} + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix}$$

$$G_t = I + F_x^T \begin{pmatrix} 0 & 0 & \frac{v_t}{\omega_t} \cos \mu_{t-1,\theta} - \frac{v_t}{\omega_t} \cos(\mu_{t-1,\theta} + \omega_t \Delta t) \\ 0 & 0 & \frac{v_t}{\omega_t} \sin \mu_{t-1,\theta} - \frac{v_t}{\omega_t} \sin(\mu_{t-1,\theta} + \omega_t \Delta t) \\ 0 & 0 & 0 \end{pmatrix} F_x$$

$$\bar{\Sigma}_t = G_t \, \Sigma_{t-1} \, G_t^T + F_x^T \, R_t \, F_x$$

$$Q_t = \begin{pmatrix} \sigma_r & 0 & 0 \\ 0 & \sigma_\phi & 0 \\ 0 & 0 & \sigma_s \end{pmatrix}$$

for all observed features $z_t^i = (r_t^i \;\; \phi_t^i \;\; s_t^i)^T$ *do*

$$\begin{pmatrix} \bar{\mu}_{N_t+1,x} \\ \bar{\mu}_{N_t+1,y} \\ \bar{\mu}_{N_t+1,s} \end{pmatrix} = \begin{pmatrix} \bar{\mu}_{t,x} \\ \bar{\mu}_{t,y} \\ s_t^i \end{pmatrix} + r_t^i \begin{pmatrix} \cos(\phi_t^i + \bar{\mu}_{t,\theta}) \\ \sin(\phi_t^i + \bar{\mu}_{t,\theta}) \\ 0 \end{pmatrix}$$

for $k = 1$ *to* $N_t + 1$ *do*

$$\delta_k = \begin{pmatrix} \delta_{k,x} \\ \delta_{k,y} \end{pmatrix} = \begin{pmatrix} \bar{\mu}_{k,x} - \bar{\mu}_{t,x} \\ \bar{\mu}_{k,y} - \bar{\mu}_{t,y} \end{pmatrix}$$

$$q_k = \delta_k^T \delta_k$$

$$\hat{z}_t^k = \begin{pmatrix} \sqrt{q_k} \\ \operatorname{atan2}(\delta_{k,y}, \delta_{k,x}) - \bar{\mu}_{t,\theta} \\ \bar{\mu}_{k,s} \end{pmatrix}$$

$$F_{x,k} = \begin{pmatrix} 1 & 0 & 0 & 0 \cdots 0 & 0 & 0 & 0 & 0 \cdots 0 \\ 0 & 1 & 0 & 0 \cdots 0 & 0 & 0 & 0 & 0 \cdots 0 \\ 0 & 0 & 1 & 0 \cdots 0 & 0 & 0 & 0 & 0 \cdots 0 \\ 0 & 0 & 0 & 0 \cdots 0 & 1 & 0 & 0 & 0 \cdots 0 \\ 0 & 0 & 0 & 0 \cdots 0 & 0 & 1 & 0 & 0 \cdots 0 \\ 0 & 0 & 0 & 0 \cdots 0 & 0 & 0 & 1 & 0 \cdots 0 \end{pmatrix}$$

$$H_t^k = \frac{1}{q_k} \begin{pmatrix} -\sqrt{q_k}\,\delta_{k,x} & -\sqrt{q_k}\,\delta_{k,y} & 0 & \sqrt{q_k}\,\delta_{k,x} & \sqrt{q_k}\,\delta_{k,y} & 0 \\ \delta_{k,y} & -\delta_{k,x} & -1 & -\delta_{k,y} & \delta_{k,x} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} F_{x,k}$$

$$\Psi_k = H_t^k \, \bar{\Sigma}_t \, [H_t^k]^T + Q_t$$

$$\pi_k = (z_t^i - \hat{z}_t^k)^T \, \Psi_k^{-1} \, (z_t^i - \hat{z}_t^k)$$

endfor

$$\pi_{N_t+1} = \alpha$$

$$j(i) = \operatorname*{argmin}_k \; \pi_k$$

$$N_t = \max\{N_t, j(i)\}$$

$$K_t^i = \bar{\Sigma}_t \, [H_t^{j(i)}]^T \Psi_{j(i)}^{-1}$$

$$\bar{\mu}_t = \bar{\mu}_t + K_t^i \, (z_t^i - \hat{z}_t^{j(i)})$$

$$\bar{\Sigma}_t = (I - K_t^i \, H_t^{j(i)}) \, \bar{\Sigma}_t$$

endfor

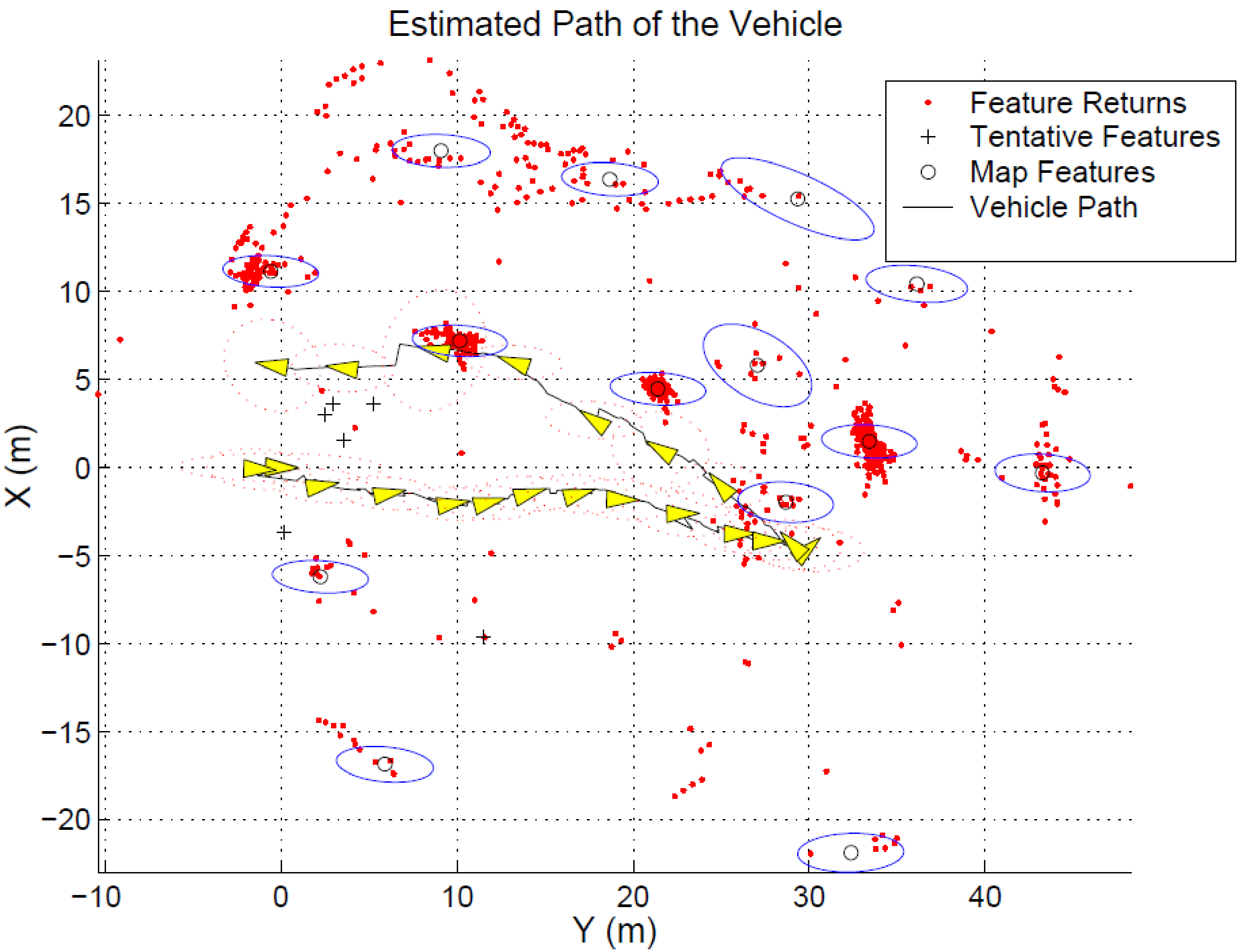
$$\mu_t = \bar{\mu}_t$$

$$\Sigma_t = \bar{\Sigma}_t$$

return μ_t, Σ_t

Unmanned Underwater Vehicle Example

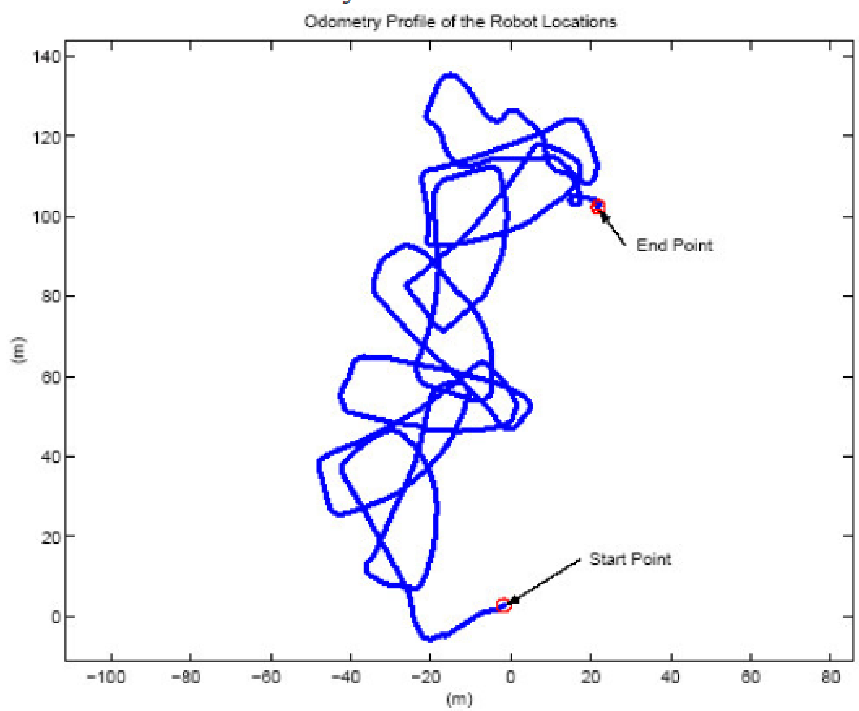
Oberon: University of Sydney Australian Center for Field Robotics



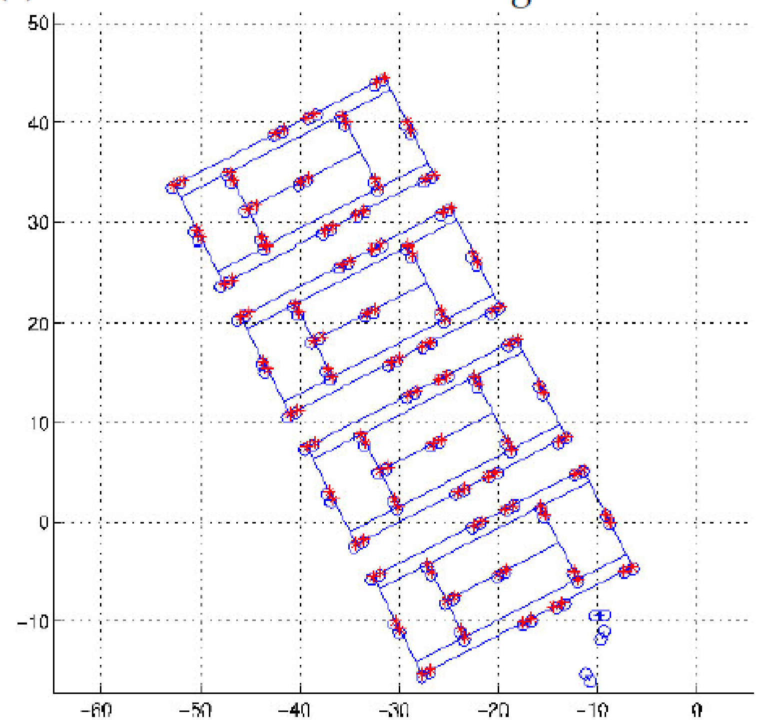
Mobile Robot: Calibrated Landmark Field

MIT B-21 Mobile Robot

(b) Raw odometry



(c) Result of EKF SLAM with ground truth



Limitations of EKF-SLAM

- **Necessity to select appropriate landmarks,**
  (Using landmarks requires pre-filtering and is a reduction in available information and ultimately is a limit to performance)
- **Quadratic update complexity limits EKF-SLAM to maps with less than about 1000 features,**
- **Small number of landmarks on the map cause harder data association**
  (Leads to the fundamental dilemma of EKF-SLAM: Maximum Likelihood data association require dense maps but feasible execution requires sparse maps!!)
- **Data association failures are usually catastrophic since EKF-SLAM cannot correct earlier data association mistakes,**
- **There are other SLAM approximations that can cope with much larger number of landmarks or raw sensor data**

Practical Issues

- **Some ideas may drastically reduce the computational complexity (but not that it is quadratic with the size of the map!!) or improve performance**
- **Outlier detection and rejection,**
- **A simple version: Filter landmarks according to robot location (Do not process those that are far away)**
- **Another one: Provisional Landmark List**
  (Keep a provisional list of landmarks that are newly seen. Keep them from updating pose until their covariance shrinks to a predetermined level)
- **Yet another: Landmark Existence Probability**
  (Keep a log existence likelihood for the landmarks. Increment when an expected landmark is seen, decrement when it is expected but missing. Delete landmark if this likelihood becomes very small.)
- **Explicit initialization of Covariance for newly seen landmarks based on multiple measurements from it.**

Approximations for SLAM

- Rao-Blackwellisation (FastSLAM)
  [Murphy 99, Montemerlo et al. 02, Eliazar et al. 03, Haehnel et al. 03]
- Local submaps
  [Leonard et al.99, Bosse et al. 02, Newman et al. 03]
- Sparse links (correlations)
  [Lu & Milios 97, Guivant & Nebot 01]
- Sparse extended information filters
  [Frese et al. 01, Thrun et al. 02]
- Thin junction tree filters
  [Paskin 03]

This will be our next focus!

EKF-SLAM Summary

- Quadratic in the number of landmarks: $O(n^2)$
- Convergence results for the linear case.
- Can diverge if nonlinearities are large!
- Have been applied successfully in large-scale environments.
- Approximations reduce the computational complexity.
- Historically the earliest and most influential version,
- There are other variants that can deal with larger problems.